

Automatic LiFePO4 Cell Charger

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Lithium iron phosphate (LiFePO₄) cell is a type of lithium-ion cell that has higher storage capacity and can deliver higher power as compared to a lead-acid cell in a lead-acid battery. LiFePO₄ batteries have long life due to their extremely robust crystal structures of the iron phosphates and have high voltage tolerances during charging and discharging.

A LiFePO₄ cell may be safely overcharged up to 4.2V and normally does not catch fire like a Li-ion or LiPo cell. These cells are being used in most leading electric vehicles as cell packs due to their higher performance and increased lifespan.

LiFePO₄ cell needs a constant-current and constant-voltage charging system, which can be fulfilled by the circuit presented here. This circuit can charge a single or two LiFePO₄ cells in parallel. The circuit charges the cell in constant-current mode up to 3.4V and then it automatically changes to constant-voltage charging mode.

Circuit and working

The circuit diagram of automatic LiFePO₄ cell charger is shown in Fig. 1. It is built around ATmega8A (IC1), 5V voltage regulator 7805 (IC2), 5V voltage regulator 7805

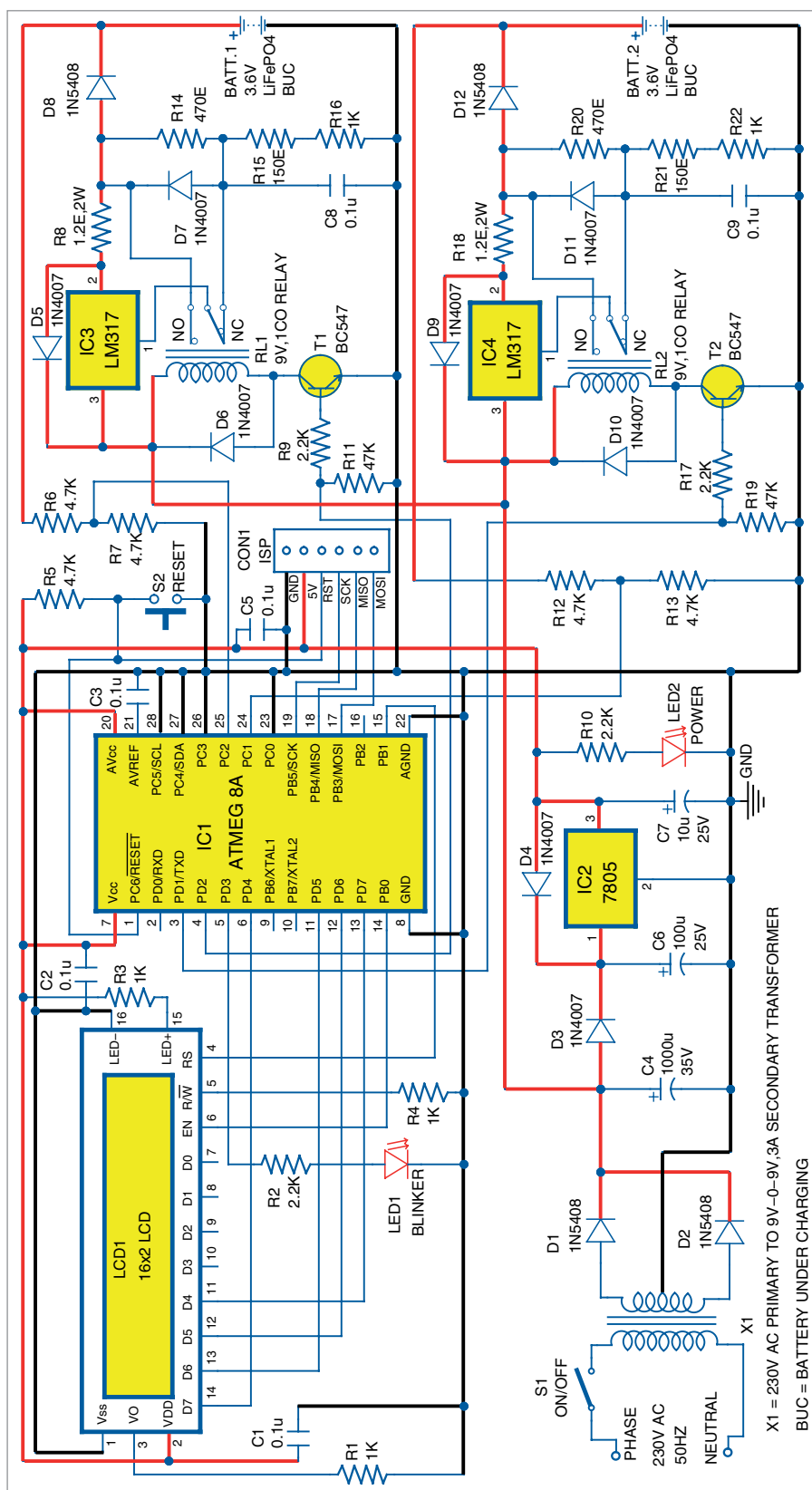


Fig. 1: Circuit diagram of the cell charger